

DAY-2 Task report

SEARCH TOPICS

1 Do's of RAG

1. Use high-quality data sources – Ensure the knowledge base contains accurate, up-to-date, and relevant information.

2. Optimize retrieval mechanisms – Use efficient search methods such as vector databases and semantic search for better results.

3. Regularly update the knowledge base – Keep documents current to improve response accuracy.

4. Validate retrieved information – Check retrieved content before generating responses to reduce errors.

5. Use proper chunking techniques – Split documents into meaningful sections to improve retrieval performance.

6. Monitor system performance – Continuously evaluate retrieval accuracy and response quality.

7. Implement security measures – Protect sensitive data and control access to information sources.

2 Don'ts of RAG

1. Don't rely on low-quality or outdated data – Poor data leads to . inaccurate responses.

- 2. Don't retrieve too much irrelevant information – Excessive context can confuse the model.***
- 3. Don't ignore data preprocessing – Unstructured or noisy data reduces retrieval effectiveness.***
- 4. Don't assume retrieval is always correct – Retrieved information may still contain errors.***
- 5. Don't neglect evaluation and testing – Regular testing is necessary to maintain quality.***
- 6. Don't expose sensitive information – Avoid retrieving or displaying confidential data.***
- 7. Don't use overly large context windows unnecessarily – This can increase costs and reduce efficiency.***

3 Limitations of RAG Systems

- 1. Dependence on retrieval quality – If relevant information is not retrieved, the generated answer may be incorrect.***
- 2. Higher latency – Retrieval and generation steps can increase response time.***
- 3. Complex system architecture – RAG requires managing databases, embeddings, and retrieval pipelines.***
- 4. Data maintenance requirements – Knowledge bases must be regularly updated and cleaned.***
- 5. Potential retrieval errors – The system may retrieve irrelevant or incomplete information.***

6. Increased computational cost – Storing embeddings and performing searches require additional resources.

7. Context limitations – Large amounts of retrieved data may exceed the model's context window.

8. Security and privacy concerns – Sensitive information in the knowledge base may be exposed if not properly protected.

SUMMARY

RAG (Retrieval-Augmented Generation) improves AI responses by combining information retrieval with language generation. To use RAG effectively, maintain high-quality data, optimize retrieval, and regularly evaluate performance. However, RAG systems still face challenges such as retrieval errors, higher costs, increased complexity, and dependency on the quality of the knowledge base.

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